



N148AV (S/N: KONUK10295956) — Disk Failure Diagnostic Report

Immfly Ground Operations / IT — Session: August 25, 2026

Field	Value
Machine	N148AV (S/N: KONUK10295956)
Disk	ADATA IM2P33E8-001TDP, S/N 2O132L1AA4RC, 1TB NVMe
Affected component	NVMe internal disk — physical read failure
Initial symptom	Suspected disk failure during SystemRescue boot for imaging
Root cause	Confirmed — critical medium error on sector 0 (boot/partition table sector), plus a second failing LBA
SMART self-assessment	PASSED (overall), but active media errors present — self-assessment does not reflect functional failure
Recommended action	Physical disk replacement required — sector 0 failure makes the disk unusable for boot/imaging regardless of overall error count

Narrative Summary

N148AV was booted into SystemRescue in preparation for restoring the N098AV golden image, following the same disk-imaging workflow already used successfully on N103AV earlier the same day. Before the restore process began, the boot log showed repeated kernel-level I/O errors immediately on startup, all pointing at logical block (sector) 0 of the internal NVMe disk — the sector that holds the partition table / boot information.

A full SMART health report was pulled from the drive (ADATA IM2P33E8-001TDP, 1TB) to confirm and quantify the issue. Notably, the drive's own overall self-assessment reports "**PASSED**" — this is a case where the summary health flag does not reflect a real, functional failure, and the detailed error log had to be examined directly to find the actual problem.

The SMART error log shows **217 total Media and Data Integrity Errors**, concentrated on two specific logical block addresses: **LBA 0** (the boot/partition-table sector, matching the kernel errors seen at boot) and **LBA 2000409248** (a second location elsewhere on the disk). Both addresses appear repeatedly across the most recent error log entries, indicating an active, ongoing failure rather than a one-time historical event.

Two additional data points stand out as context: the drive has only **104 power-on hours** total, making it a relatively young unit in terms of usage, yet it has already accumulated **29 unsafe shutdowns out of 38 total power cycles (76%)** — a disproportionately high rate of abrupt power loss events relative to its age, which may be a contributing factor to the current media degradation. "Percentage Used" (the drive's own wear-leveling/endurance indicator) reads 0%, confirming this is not a simple end-of-life wear issue but rather localized physical media damage.

Because LBA 0 specifically fails to read, the disk cannot reliably present its partition table to the OS or bootloader — this makes the drive unusable for booting or for the disk-imaging restore process, regardless of the fact that the raw error count is moderate compared to other units previously diagnosed in the fleet. The restore process for N148AV was halted before writing any data to avoid attempting to partition or write over an already-failing sector 0.

Diagnostic Steps Performed

1. Boot into SystemRescue

N148AV was booted from the SystemRescue rescue disk in preparation for restoring the N098AV golden image. Kernel boot log immediately showed repeated 'critical medium error' and 'Buffer I/O error' messages on dev nvme0n1, sector 0, op READ.

2. Disk detection check

lsblk confirmed the NVMe disk is detected at the bus level (953.9G, visible as nvme0n1), ruling out a total bus-level failure like the one previously seen on N158AV — this is a more localized, read-specific failure rather than complete disk absence.

3. Full SMART health report

smartctl -a /dev/nvme0n1 pulled complete SMART/NVMe health data. Overall self-assessment reports PASSED, but the detailed error log shows 217 Media and Data Integrity Errors, with the most recent 16 logged entries repeatedly pointing to LBA 0 and LBA 2000409248.

4. Restore halted

Given the confirmed active failure on the boot/partition-table sector, the restore process was not attempted on this disk to avoid further complications or wasted time on a drive that cannot reliably hold a bootable image.

Conclusion and Recommendation

N148AV's internal NVMe disk (ADATA IM2P33E8-001TDP, S/N 20132L1AA4RC) has a confirmed, active hardware failure affecting sector 0 (the boot/partition table sector) as well as a second logical block elsewhere on the disk. This is a functional failure that prevents the disk from being reliably imaged, partitioned, or booted, despite the drive's own SMART self-assessment reporting "PASSED" — the summary flag should not be relied upon alone for drives showing active error log entries.

Recommended action: Physical disk replacement is required before N148AV can be restored to service. Once a replacement NVMe disk is installed, the standard N098AV golden-image restore workflow (already validated successfully on N103AV) can proceed normally. Given the disproportionately high unsafe-shutdown ratio (76%) relative to the drive's low power-on hours, it may also be worth investigating whether N148AV has an underlying power-delivery issue independent of the disk itself, to prevent a replacement drive from degrading the same way.

Appendix: Raw SMART Output

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Model Number: ADATA_IM2P33E8-001TDP
Serial Number: 20132L1AA4RC
Firmware Version: 42BFU4MB
Namespace 1 Size/Capacity: 1,024,209,543,168 [1.02 TB]
Namespace 1 Utilization: 274,229,813,248 [274 GB]

SMART overall-health self-assessment test result: PASSED

Critical Warning: 0x00
Temperature: 30 Celsius
Available Spare: 100%
Available Spare Threshold: 10%
Percentage Used: 0%
Data Units Read: 67,239 [34.4 GB]
Data Units Written: 598,148 [306 GB]
Power Cycles: 38
Power On Hours: 104
Unsafe Shutdowns: 29
Media and Data Integrity Errors: 217
Error Information Log Entries: 217

Error Information (most recent entries, LBA column):
LBA 2000409248 (7 of 16 recent entries)
LBA 0 (6 of 16 recent entries) -- boot/partition table sector

Kernel boot log:
blk_update_request: critical medium error, dev nvme0n1, sector 0 op 0x0:(READ)
Buffer I/O error on dev nvme0n1, logical block 0, async page read
(repeated 5+ times during boot)
```